

Pattern Recognition Letters

GRA-CNN: A Novel Convolutional Neural Network Method with Gray Relational Analysis

--Manuscript Draft--

Manuscript Number:	PRLETTERS-D-25-01152
Article Type:	Regular Article
Keywords:	Deep Learning, Convolutional Neural Network, Gray Relational Analysis, Image Recognition.
Corresponding Author:	Kangrui Li Guangdong University of Technology CHINA
First Author:	Xiaobo Zhang
Order of Authors:	Xiaobo Zhang
	Kangrui Li
	Junyi Lin
	Yuting Wen
Abstract:	Deep convolutional neural network is one of the most important deep learning models, widely used in image recognition, speech recognition and other fields. In the research and application of convolutional neural networks, the number of hidden layers and nodes is generally determined by experience and expert knowledge. In this sense, achieving optimal network architecture with excellent generalization capabilities and low model complexity is time consuming and even impossible. In order to solve this problem, this paper proposes a gray correlation analysis method to optimize the convolutional neural network (GRA-CNN). The method evaluates the correlation between hidden nodes and output nodes in the network model and selects hidden nodes with higher correlation. Deleting hidden nodes with smaller degrees ensures that the final convolutional neural network can have better recognition accuracy while maintaining higher training performance and higher system stability and adaptability. Experiments on several benchmark datasets show that the proposed method is superior to many other methods in terms of model performance or model complexity.

Confirmation of Authorship

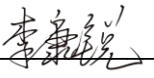
Pattern Recognition Letters

Authorship Confirmation

Please save a copy of this file, complete and upload as the “Confirmation of Authorship” file.

As corresponding author I, Kangrui Li, hereby confirm on behalf of all authors that:

1. This manuscript, or a large part of it, has not been published, was not, and is not being submitted to any other journal.
2. If presented at or submitted to or published at a conference(s), the conference(s) is (are) identified and substantial justification for re-publication is presented below. A copy of conference paper(s) is(are) uploaded with the manuscript.
3. If the manuscript appears as a preprint anywhere on the web, e.g. arXiv, etc., it is identified below. The preprint should include a statement that the paper is under consideration at Pattern Recognition Letters.
4. All text and graphics, except for those marked with sources, are original works of the authors, and all necessary permissions for publication were secured prior to submission of the manuscript.
5. All authors each made a significant contribution to the research reported and have read and approved the submitted manuscript.

Signature  (Kangrui Li) Date 3 November 2025

List any pre-prints:

None

Relevant Conference publication(s) (submitted, accepted, or published):

None

Justification for re-publication:

Not applicable

Cover Letter

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “**GRA-CNN: A Novel Convolutional Neural Network Method with Gray Relational Analysis**” for consideration for publication in *Pattern Recognition Letters*.

This paper proposes a convolutional neural network optimization framework integrating Gray Relational Analysis (GRA) for adaptive pruning and feature correlation evaluation. The proposed GRA-CNN model improves recognition accuracy and computational efficiency by reducing redundant hidden nodes. Experimental validation on benchmark datasets demonstrates the superiority of GRA-CNN over conventional CNNs in terms of accuracy, stability, and model compactness.

This manuscript has not been published or submitted elsewhere, and all authors have approved the submission. We confirm that all figures and tables are original and that no part of this work has been published previously.

We believe that our work fits well within the scope of *Pattern Recognition Letters*, contributing to the understanding of deep learning model optimization and gray relational methods in pattern recognition.

Thank you for considering our submission.

Sincerely,

Kangrui Li* (Corresponding Author)

Faculty of Automation, Guangdong University of Technology
Guangzhou, China

E-mail: 1403073295@mails.gdut.edu.cn



GRA-CNN: A Novel Convolutional Neural Network Method with Gray Relational Analysis

Xiaobo Zhang ^a, Kangrui Li ^{a,*}, Junyi Lin ^a, Yuting Wen ^b

^aFaculty of Automation, Guangdong University of Technology, Guangzhou, 510006, China

^bGuangzhou Goaland Energy Conservation Tech.Co.,Ltd.,510663, China

ABSTRACT

Deep convolutional neural network is one of the most important deep learning models, widely used in image recognition, speech recognition and other fields. In the research and application of convolutional neural networks, the number of hidden layers and nodes is generally determined by experience and expert knowledge. In this sense, achieving optimal network architecture with excellent generalization capabilities and low model complexity is time consuming and even impossible. In order to solve this problem, this paper proposes a gray correlation analysis method to optimize the convolutional neural network (GRA-CNN). The method evaluates the correlation between hidden nodes and output nodes in the network model and selects hidden nodes with higher correlation. Deleting hidden nodes with smaller degrees ensures that the final convolutional neural network can have better recognition accuracy while maintaining higher training performance and higher system stability and adaptability. Experiments on several benchmark datasets show that the proposed method is superior to many other methods in terms of model performance or model complexity.

Keywords: Deep Learning, Convolutional Neural Network, Gray Relational Analysis, Image Recognition.

* Corresponding author: 1403073295@mails.gdut.edu.cn(K. Li)

E-mails: zxb_leng@gdut.edu.cn(X. Zhang), 3123001378@mail2.gdut.edu.cn(J. Lin), 3223001408@mail2.gdut.edu.cn(Y. Wen).

1. Introduction

Deep learning is an active and exciting branch in machine learning and artificial intelligence research [1], [2]. The intuition of deep learning originates from the biological research of human brains. Deep network simulates the connection and activation of nerve cells in a brain. By significantly increasing the layers between input and output units, deep models obtain the ability of learning complexity functions, recognizing abstract concepts and processing many intelligent tasks. Deep learning models can learn essential characteristics from massive data, sparking innovations across medicine and finance [3], [4]. The successful applications of deep learning models are now changing people's lives and making our world increasingly intelligent [5]-[7]. A systematic comparison between conventional feature-engineering pipelines and modern deep learning approaches further clarifies this transition [6].

On the other hand, since images are widely used in information transmission and knowledge sharing through internet, classification, recognition or grading become great challenge when building intelligent image processing applications. The low training and testing cost of the model are preferred especially in real-time and customer-friendly applications. The process of image recognition includes image preprocessing, segmentation, feature extraction and model training [8]. With the help of image recognition and classification, the fast retrieval of image information can be achieved that has a widely range of application such as face recognition [9], [10] safety protection [11], autonomous driving [12], and medical image [13] and so on.

The traditional image recognition model usually adopts a domain-specific feature extractor and a shallow model for recognition. Figure 1 sketches the main steps of a traditional image recognition model.

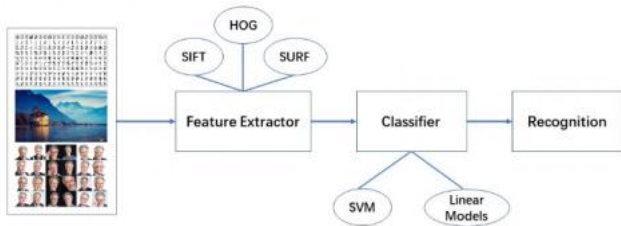


Fig. 1. Main steps of traditional image recognition model

Roughly speaking, traditional image recognition model generates clusters of similar images in the feature space induced by the feature representation, and achieves image recognition by classifiers such as KNN and SVM [14]-[15]. For example, HOG features are widely used in pedestrian detection, LBP features in face recognition, and SIFT features in object recognition, etc.

The main steps of image recognition based on deep learning are as follows. Raw pixels of a test image are directly fed to the deep network model. In the training stage, the network parameters, i.e. weights and biases, are updated according to the back-propagated training errors and the training rate. When the evaluation errors tend to be stable, the model converges and can be used as a recognition model [16]. Figure 2 shows the main steps of the deep learning-based image recognition model.

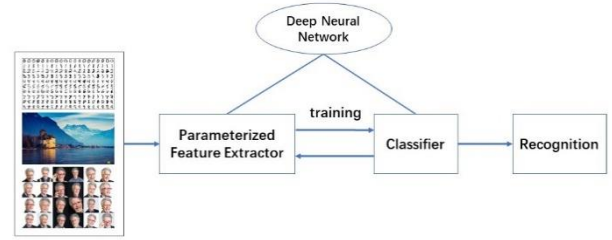


Fig. 2. Main steps of the deep learning-based image recognition model

Commonly, deep learning-based image recognition system includes feature extraction module and image recognition module. To achieve good performance, these two modules should be jointly trained and optimized. In this paper, we proposed an image recognition model based on convolutional neural network using gray relational analysis method (GRA-CNN).

2. Related Works

Machine learning models can be divided into shallow model and deep level. Traditional machine learning model usually adopts a shallow structure with 2-3 layers. For example, Gaussian mixture model, K-means clustering, support vector machine are representative of traditional machine learning [17]. These shallow models can achieve good results in solving simple practical problems, but they are not effective in expressing complex real-world problems [18].

From a narrow perspective, deep learning is a neural network that contains several hidden layers and has certain training methods and structures [19]. From a broad perspective, it is well accepted that any machine learning model with a deep hierarchical structure can be regarded as a deep learning model [20]. In the process of deep learning, the unsupervised learning method is used to process the input image and learn the features of the image. Then, the supervising sequence is used to update the entire network structure parameters. Finally, the classification result of the image is output at the output layer. In recent years, with the rapid development of deep learning, it has been widely used in voice recognition, image classification, natural language processing and other fields.

In 1974, Paul Werbos proposed a back-propagation (BP) algorithm to solve the linear inseparable problem of neural network applied to deep layers. However, when the number of network layers increase, BP neural network will also encounter problems such as "gradient disappearance" or instability of training structure, which leads to the inability of deep neural network [21]. A famous deep model called Deep Belief Network (DBN) was proposed and successfully applied to many classification tasks. A typical DBN model has a deep belief network by stacking several Restricted Boltzmann Machines (RBM), and solved the problems of gradient disappearance and local optimization by unsupervised layer-wise greedy algorithms [22]. Vincent et al. proposed the deep structure of the denoising Stacked Automatic Encoder (SAE) [23] to further improve the accuracy of unsupervised pre-training. LeCun proposed a Convolutional Neural Network (CNN) based on DBN and SAE [24]. DBN, SAE and CNN constitute the three fundamental frameworks for deep learning and have made great achievements in natural image processing and face recognition.

With the evolution of CNN architectures such as VGG net [20], deeper and more uniform convolutional layers demonstrated that increasing network depth can significantly improve image recognition performance, laying the foundation for later architecture optimization.

3. The Proposed GRA-CNN Method

3.1. Challenges of convolutional neural network construction

The number of hidden nodes is one of the most important factors for the network structure. The hidden feature map determines the number of hidden nodes in the network. The less hidden feature maps are the less information network obtains. The lower accuracy is the slower convergence rate is. On the contrary, the more hidden feature maps are, the higher accuracy of network is, but it maybe leads to training time too long and overtraining that result in worse recognition.

In the process of research and application of convolutional neural network, the hidden nodes are generally obtained by experience. By constructing and training a network structure that contains different numbers of hidden nodes, the best network structure is selected as the final network structure. This method of constructing a network structure has limited capability for it is difficult to find an optimal network structure. In addition, in the convolutional neural network, the influence of different feature graph units on the output layer nodes is different, some of the feature graph units have a greater impact on the output nodes, and the other part of the feature graph units have a smaller one. Therefore, it is possible to improve the performance of convolutional neural network by deleting the unit with small influence without affecting the training effect. Moreover, improvements on activation behavior—such as introducing sparse regularization for ReLU—have been shown to enhance generalization and convergence stability [25].

3.2. The structure optimization of CNN

In view of the difficulty of convolutional neural network, this paper uses gray relational analysis to determine the number of feature graphs. The key of this process is to use the gray relational analysis method to calculate the correlation degree between hidden nodes and output nodes, and sort these hidden nodes to retain those with high correlation degree. The hidden nodes with smaller degrees are deleted to ensure the final network structure can have better recognition accuracy while maintaining higher training performance and higher system stability and adaptability.

The application method of gray relational analysis to improve convolutional neural network is shown in Figure 3.

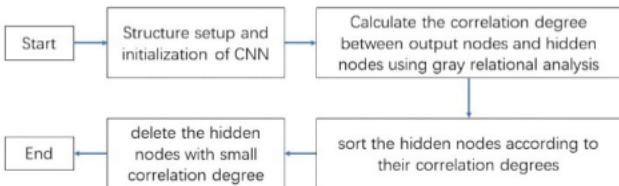


Fig. 3. The improved flow chart of convolutional neural network based on gray relational analysis

In this paper, the structure of convolutional neural network is shown in Figure 3. In the process of constructing the convolutional neural network, the number of nodes in each hidden layer is calculated by the gray correlation method. Taking the S4 layer as an example, the improvement process is as follows.

1) Correlation coefficient calculation

The initial convolutional neural network is used to process a batch of sample images and the result of the sample image processing and the identification data in the process are obtained. At the same time, a weight matrix of the output value from the S4 layer to the output layer is extracted from the identification data, and is set as a database. And then use the gray correlation method to calculate the output of each feature map in the S4 layer and its correlation coefficient with the output of the entire network.

It is assumed that there are a total of N sample images to be processed in the process of convolutional neural network optimization, and the number of feature maps in S4 layer is n and $y = (y_1, y_2, \dots, y_N)$ is the output of the network (referred to as a reference sequence). The output of each feature map of S4 layer is denoted by $t_i = (t_i(1), t_i(2), \dots, t_i(N))$, $i = 1, 2, \dots, N$ (called the comparison sequence). The formula of the correlation coefficient between network output y and the output of feature map is shown in Eq. (1).

$$\xi_i(k) = \frac{\min_i \min_k |y_k - t_i(k)| + \rho \max_i \max_k |y_k - t_i(k)|}{|y_k - t_i(k)| + \rho \max_i \max_k |y_k - t_i(k)|} \quad (1)$$

$\rho (\rho \in [0, 1])$ is the resolution coefficient in the process of calculating the correlation coefficient and its default value is set to 0.5.

2) Correlation degree calculation

The correlation coefficient is the correlation degree between the reference sequence and the comparison sequence under different image samples. For the N sample images to be processed there are N correlation coefficients. In order to analyze, it is necessary to transfer N correlation coefficients into a value such as correlation degree. The formula of correlation degree calculation is shown in Eq. (2).

$$y_i = \frac{1}{N} \sum_{k=1}^N \xi_i(k) \quad (2)$$

3) Ranking of correlation degree

After the processing of N sample images, the correlation degree of all the hidden features of the network is obtained, and then all the hidden features are arranged in descending order and a threshold value is set as $\mathcal{E}$, the feature maps smaller than the threshold value are omitted. Through this method convolutional neural network could be optimized and the performance could be improved.

4. Experiment And Evaluation

Based on the optimization of convolutional neural network, taking the road traffic sign as an example, the experiment of the optimized convolutional neural network in image recognition is studied.

4.1. Evaluation Dataset Description

The German Traffic Sign Recognition Benchmark (GTSRB) dataset [26] is used to evaluate the proposed method. There are totally 43 categories of traffic signs in the GTSRB. The dataset has been divided into training set and test set, whose sizes are 39209 and 12630. As is shown in Figure 4.



Fig 4. Selected 10 types of traffic sign images

This data set is used for experiments because it is a publicly available data set for researchers to experiment with. The data set is relatively in line with international standards. Using this data set to verify our method can improve the standard and generality of our method.

After performing the following preprocessing, the selected 6,000 sample images are used as the graphic database for this convolutional neural network experiment.

1) Binarization

Traffic sign images in GTSRB image database are shot randomly. In order to facilitate the processing, it is necessary to binarize the whole dataset for efficient information extraction and image recognition.

The binarization method involves setting a threshold value, setting pixels in the image that are less than the threshold to 0, and setting the others to 1.

2) Uniform pixel size

The image samples to be processed are stretched and compressed so that the 6000 samples to be processed have the same pixel row and column size to facilitate the experiment.

4.2. Settings of Network Structure And Parameters

The setting of all parameters in this experiment uses universal parameter settings to verify our proposed method. This can prove as much as possible that our proposed method has the advantages and versatility, and reduces special cases.

1) Network structure layer and convolution kernel size setting

Set the number of convolutional neural network layer as shown in Figure 2 to 4, set the convolution kernel size to 5×5 , set the size of input image to 28×28 , set the size of output feature map passing through the C1 convolution layer to 24×24 , set the size of output feature map passing through the S2 sampling layer to 12×12 , set the size of output feature map passing through the C3 convolution layer to 8×8 , and set the size of output feature map passing through the S4 sampling layer to 4×4 .

2) Batch size setting

As mentioned above, the training sample for the convolutional neural network experiment is 39209, the test sample is 12630. To clearly show the impact of batch size, we set the batch size to the following values: 10, 20, 40, 50, 100, 250 and 500. The error rates and experimental run times obtained by iterating different times are shown in Figure 5 and Figure 6 respectively.

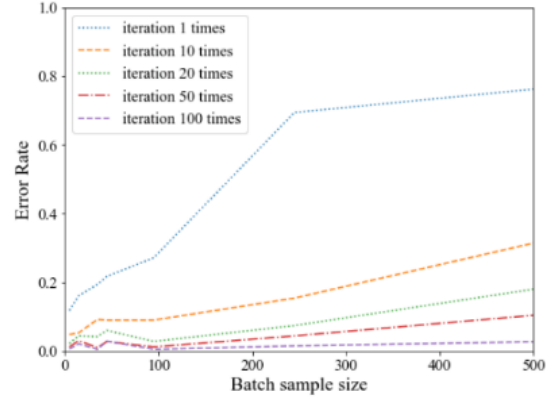


Fig. 5. Batch training results of image samples

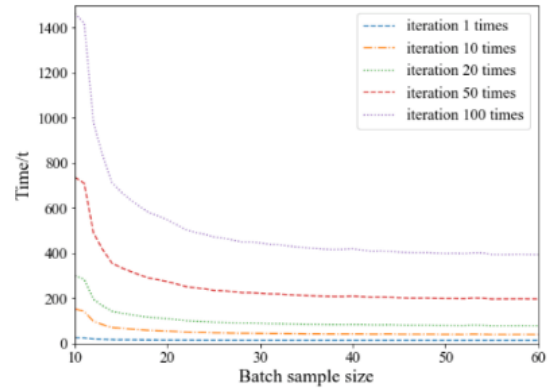


Fig. 6. Run time results of different batches of samples

As indicated in Figure 5 and Figure 6, the batch size is smaller; the time to get a higher accurate rate is longer. According to the above error rate and operational performance, the number of batch samples determined in this experiment is 40.

3) Rounds of iteration

In this evaluation, the batch size is set to 40. The experimental results for the iteration times are shown in Figure 7.

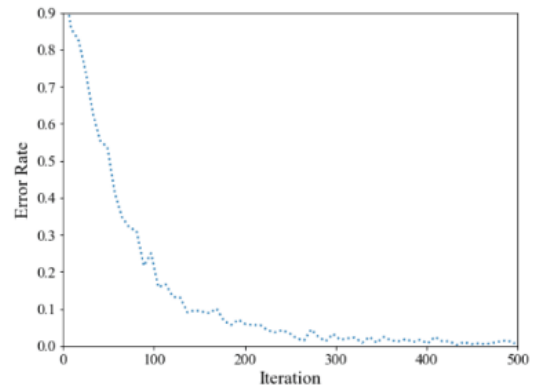


Fig. 7. Experimental results for iteration times

As shown in Figure 7, more rounds of iterations mean better recognition performance, but the training time would increase accordingly. We leverage the model performance and the computational cost by setting the number of rounds to 300.

4) Number setting of hidden feature map

This paper chooses the convolutional neural network structure as shown in Figure 1, including two convolution layers and two sampling layers. The feature maps number of the sampling layer is determined by the number of the convolution layer. The feature maps number of convolution layer is generally determined experimentally, and the feature maps number of the C1 convolution layer is kept unchanged. By changing the feature maps number of C3 convolution layer, we can observe the change of error rate and select the optimal network structure.

During the experiment, the value of C1 convolution layer is [1, 10], the C3 convolution layer is [1, 40]. With the value test, we select the value combinations of lowest error rate and better time cost. The results are shown in Figure 8 and Figure 9.

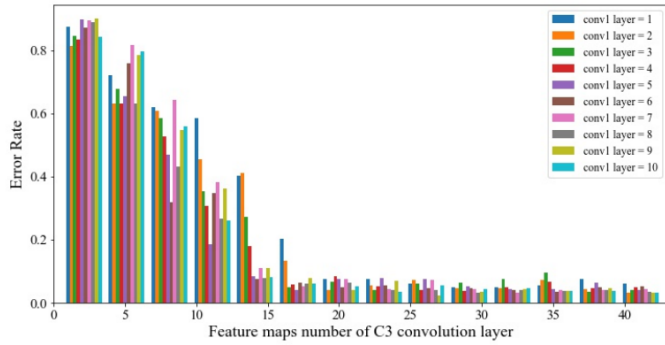


Fig. 8. The experimental results of convolution layer feature maps with error rate

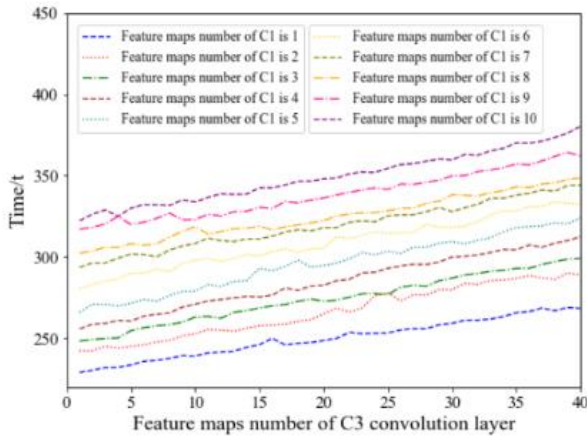


Fig. 9. The experimental results of convolution layer feature maps with time cost

As shown in Figure 8 and Figure 9, the feature maps for C1 convolution layer are selected as 8 and C3 convolution layer as 18, the network image recognition error rate begins to converge to around 5% and has good performance in terms of time cost.

4.3. Optimization Of Network Structure

This experiment uses the aforementioned optimization method to determine the number of hidden layer features by the following three parts: First, the number of S4 layer feature maps is determined according to the correlation degree between the S4 layer and the output layer. Then, according to the correlation degree between the S2 layer and the C3 layer, the number of S2 layer feature maps is determined. Finally, we perform experiment on the accuracy of the network structure.

1) Determine the feature map numbers of C3, S4 layer

The initial structure is set as 1-10-30-10, the resolution coefficient p is set to 0.5 and by the training of the network, and

the correlation rank of C3 and S4 layers is obtained as shown in Table 1.

Table 1. Related Correlation Of S4 Hidden Layer Features

No.	Corr.	No.	Corr.	No.	Corr.
1	0.5791	2	0.5770	3	0.5747
4	0.5721	5	0.5544	6	0.5509
7	0.5494	8	0.5445	9	0.5349
10	0.5319	11	0.5254	12	0.5237
13	0.5213	14	0.5185	15	0.5181
16	0.5164	17	0.5158	18	0.5142
19	0.5139	20	0.5105	21	0.5100
22	0.5091	23	0.5090	24	0.5017
25	0.5014	26	0.4983	27	0.4977
28	0.4946	29	0.4942	30	0.4934

The threshold ε is set to 0.514 by average, select the first 18 features of Table 1, and adjust the network structure as 1-10-18-10, and then train the adjusted network. The training results are shown in Table 2.

Table 2. Related Correlation Of Adjusted S4 Hidden Layer Features

No.	Corr.	No.	Corr.	No.	Corr.
1	0.6464	2	0.6328	3	0.6252
4	0.6147	5	0.6128	6	0.6057
7	0.6044	8	0.5794	9	0.5769
10	0.5747	11	0.5671	12	0.5641
13	0.5601	14	0.5526	15	0.5468
16	0.5344	17	0.5275	18	0.5214

As shown in Table 2, the 18 selected feature maps have greater impact on the result. That is, the adjustment for network structure is reasonable.

2) Determine the number of feature maps of C1 and S2 layers

After determining the number of feature maps of C3 and S4, the number of feature maps of C1 and S2 layers is determined in the same way. The ranking of the first C1 and S2 correlations is as shown in Table 3.

Table 3. Related Correlation Of S2 Hidden Layer Feature

No.	Corr.	No.	Corr.	No.	Corr.
1	0.6665	2	0.6650	3	0.6634
4	0.6439	5	0.6418	6	0.6391
7	0.6340	8	0.6339	9	0.6228
10	0.6045				

The threshold ε is set to 0.630 by average and the first 8 features of Table 3 adjust the network structure as 1-8-18-10 and train the adjusted network. The training results are shown in Table 4.

Table 4. Related Correlation Of Adjusted S2 Hidden Layer Features

No.	Corr.	No.	Corr.	No.	Corr.
1	0.6813	2	0.6722	3	0.6619
4	0.6592	5	0.6499	6	0.6487
7	0.6402	8	0.6355		

As shown in Table 4, when the number of S2 feature maps is 8, each feature map has a significant impact on the result, so the final adjusted network structure is 1-8-18-10.

4.4. Testing For Optimized Convolutional Neural Network

1) Test process

The entire testing process is divided into three steps: network structure training, feature extraction and network testing.

(1) Network structure training

The process of training is as follows:

a. According to different traffic sign image category, set the classification label.

b. Initialization of network parameters and learning rate. According to the above study, the number of layers of convolutional neural network is set as 4(1-8-18-10), and the number of batch samples is set to 40 and the number of iterations is set to 60. Using Grey Relational Analysis to determine the number of hidden feature maps.

c. Take batch samples from the graphic experiment library and train the batch samples using the convolutional neural network to obtain the entire actual output vector and the output vector of the middle layer.

d. Calculate the output error of the entire network and the output error of the hidden layer.

e. Calculate the adjustment ΔW and the bias Δb of the weight.

f. Update bias and weight of convolutional neural network.

g. Complete this iteration. If the total number of iterations reaches 60, then the training ends up, or else go to step C, continue the next iteration.

(2) Feature extraction

Taking the traffic sign of "speed limit" as an example, the process of extracting implicit features of convolutional neural network is described, as is shown in Figure 10.

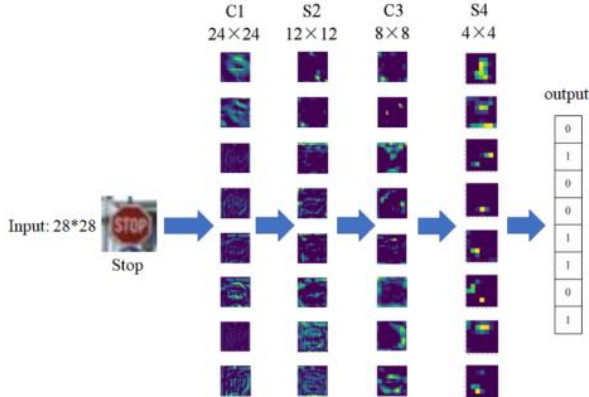


Fig. 10. Feature extraction of layers

The sample image as input data, propagated layer by layer until the output results. After the input image samples passed

through 8 filters in the C1 convolution layer, 8 feature images are formed, and after 2×2 down-sampling, the S2 layer is obtained. Then, by passing the C3 convolution layer with 18 kernels, $18 \times 4 \times 4$ images are obtained in the S4 layer and propagated to the output layer of the 10×1 convolutional neural network by the fully connected method.

(3) Network testing

After training by 5000 image samples, a trained convolutional neural network structure is obtained. Now we select the remaining 1000 image samples from the graphic experiment library to test the network structure. The test steps are as follows:

a. Select the image samples to be tested, and perform image recognition by trained convolutional neural network.

b. According to the recognition result, compare with the real label of the image sample.

c. After testing the 1000 image samples, perform statistical accuracy and error rates.

2) Evaluation results

First, according to the above experimental procedure, we use a convolutional neural network with a network structure of 1-8-30-10 to train and test. During the experiment, the number of feature maps of C1 and S2 remained the same, while the number of feature maps of C3 and S4 increased from 1 to 30, and the error rates and the time spent in 60 iterations were calculated respectively for different feature maps. The test results are shown in Figure 11.

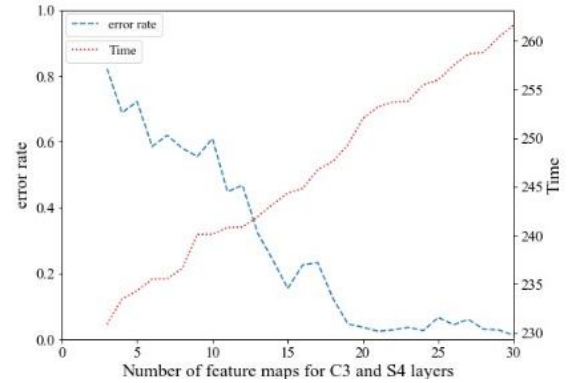


Fig. 11. Runtime and error rate of the test

As can be seen from the experimental results in Figure 11, when the number of feature maps in C3 and S4 layers is too large, the error rate will increase and the iteration will take too long. When the number of features maps in C3 and S4 layers is too small, the error rate will also be high. After adopting the gray relational analysis method to optimize the structure, the convolutional neural network structure is finally determined as 1-8-30-10.

All of the above shows that using gray relational analysis to optimize the hidden layer feature map of convolutional neural network, a convolutional neural network structure with small error rate and short running time can be obtained. Therefore, it is very significant for improving the accuracy and efficiency of image recognition to use the gray relational analysis method to optimize the convolutional neural network.

5. Conclusion

Image classification is an important field in pattern recognition. It is necessary to find an image classification method that can be used universally and has good performance on

different classification problems. At present, the image classification methods are mainly to extract the image information by manual feature extraction method. For different image classification problems, we need use different feature extraction methods, and the most effective method of a certain class is not necessarily valid on other issues. The emergence of convolutional neural network brings a new research idea to image classification. It adopts local experience in structure, weight sharing and down-sampling technology to take full account of the translation, rotation and zoom invariance of the image in space. Even if the attitude and shape of the object have changed, the recognition task can still be completed. In addition, it combines the feature extraction and classification of the traditional image classification technology, and the image is classified by using the hierarchical structure to gradually abstract the implicit features of the original image.

References

- [1] Mathew, A., Amudha, P., Sivakumari, S., 2021. Deep learning techniques: an overview, in: Satapathy, S.C., Bhateja, V., Mohanty, J.R. (Eds.), *Advances in Machine Learning and Computational Intelligence (AMLTA 2020)*. Springer, Singapore, pp. 599–608.
- [2] Bhatt, D., Patel, C., Talsania, H., Patel, J., Vaghela, R., Pandya, S., Modi, K., Ghayvat, H., 2021. CNN variants for computer vision: history, architecture, application, challenges and future scope. *Electronics* 10(20), 2470.
- [3] Jiang, W., 2021. Applications of deep learning in stock market prediction: recent progress. *Expert Systems with Applications* 184, 115537.
- [4] Gheisari, F., Wang, G., Shirmohammadi, A., Shah, S.L., Aghaei, M., 2023. Deep learning: applications, architectures, models, tools, and frameworks: a comprehensive survey. *CAAI Transactions on Intelligence Technology* 8(3), 581–606.
- [5] Sit, M., Demiray, B.Z., Xiang, Z., Ewing, G.J., Sermet, Y., Demir, I., 2020. A comprehensive review of deep learning applications in hydrology and water resources. *Water Science and Technology* 82(12), 2635–2670.
- [6] O'Mahony, N., Campbell, S., Carvalho, A., Harapanahalli, S., Hernandez, G.V., Krpalkova, L., Riordan, D., Walsh, J., 2019. Deep learning vs. traditional computer vision, in: Arai, K., Kapoor, S., Bhatia, R. (Eds.), *Advances in Computer Vision*. Springer, Cham, pp. 128–144.
- [7] Minaee, S., Kalchbrenner, N., Cambria, E., Nikzad, N., Chenaghlu, M., Socher, R., 2021. Deep learning-based text classification: a comprehensive review. *ACM Computing Surveys* 54(3), 1–40.
- [8] Minaee, S., Boykov, Y., Porikli, F., Plaza, A.J., Kehtarnavaz, N., Terzopoulos, D., 2022. Image segmentation using deep learning: a survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44(7), 3523–3542.
- [9] Li, L., Mu, X., Li, S., Peng, H., 2020. A review of face recognition technology. *IEEE Access* 8, 139110–139120.
- [10] Kortli, Y., Jridi, M., Al Falou, A., Atri, M., 2020. Face recognition systems: a survey. *Sensors* 20(2), 342.
- [11] Nath, N.D., Behzadan, A.H., Paal, S.G., 2020. Deep learning for site safety: real-time detection of personal protective equipment. *Automation in Construction* 112, 103085.
- [12] Grigorescu, S., Trasnea, B., Cocias, T., Macesanu, G., 2020. A survey of deep learning techniques for autonomous driving. *Journal of Field Robotics* 37(3), 362–386.
- [13] Cai, L., Gao, J., Zhao, D., 2020. A review of the application of deep learning in medical image classification and segmentation. *Annals of Translational Medicine* 8(11), 713.
- [14] Wang, P., Fan, E., Wang, P., 2021. Comparative analysis of image classification algorithms based on traditional machine learning and deep learning. *Pattern Recognition Letters* 141, 61–67.
- [15] Khan, A.A., Laghari, A.A., Awan, S.A., 2021. Machine learning in computer vision: a review. *EAI Endorsed Transactions on Scalable Information Systems* 8(32), e4.
- [16] Hinton, G.E., Salakhutdinov, R.R., 2006. Reducing the dimensionality of data with neural networks. *Science* 313(5786), 504–507.
- [17] Sinaga, K.P., Yang, M.S., 2020. Unsupervised K-means clustering algorithm. *IEEE Access* 8, 80716–80727.
- [18] Sarker, I.H., 2021. Machine learning: algorithms, real-world applications and research directions. *SN Computer Science* 2(3), 160.
- [19] Taye, M.M., 2023. Understanding of machine learning with deep learning: architectures, workflow, applications and future directions. *Computers* 12(5), 91.
- [20] Simonyan, K., Zisserman, A., 2015. Very deep convolutional networks for large-scale image recognition, in: *International Conference on Learning Representations (ICLR 2015)*. San Diego, USA.
- [21] Bishop, C.M., 2006. *Pattern recognition and machine learning*, first ed. Springer, New York.
- [22] Hua, Y., Guo, Z., Zhao, H., 2015. Deep belief networks and deep learning, in: *2015 International Conference on Intelligent Computing and Internet of Things (ICIT)*. IEEE, Harbin, China, pp. 1–4.
- [23] Vincent, P., Larochelle, H., Bengio, Y., Manzagol, P.A., 2008. Extracting and composing robust features with denoising autoencoders, in: *Proceedings of the 25th International Conference on Machine Learning (ICML '08)*. ACM, Helsinki, Finland, pp. 1096–1103.
- [24] Sercu, T., Puhersch, C., Kingsbury, B., LeCun, Y., 2016. Very deep multilingual convolutional neural networks for LVCSR, in: *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Shanghai, China, pp. 4955–4959.
- [25] Ide, H., Kurita, T., 2017. Improvement of learning for CNN with ReLU activation by sparse regularization, in: *2017 International Joint Conference on Neural Networks (IJCNN)*. IEEE, Anchorage, USA, pp. 2684–2691.
- [26] Stallkamp, J., Schlipsing, M., Salmen, J., Igel, C., 2012. Man vs. computer: benchmarking machine learning algorithms for traffic sign recognition. *Neural Networks* 32, 323–332.

Highlights

- Introduces GRA-CNN, a CNN enhanced with Gray Relational Analysis.
- Improves understanding of correlations between hidden and output layers.
- Dynamically prunes low-correlation nodes to boost efficiency.
- Achieves higher accuracy and robustness on benchmark datasets.
- Outperforms conventional CNN baselines in accuracy and compactness.

Declaration of Interest Statement

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: